MIL-R-46509A(MI)
22 OCTOBER 1982
SUPERSEDING
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17 March 1961

MILITARY SPECIFICATION

* ROCKET MOTOR, M30, METAL PARTS FOR

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE .

* 1.1 SCOPE. This specification covers requirements for the metal parts for one type of rocket motor.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-W-423	Wire, Steel, Corrosion-Resisting
* TT-E-516	Enamel, Lusterless, Quick-Drying, Styrenated Alkyd Type

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-RSDS, Redstone Arsenal, AL 35898, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1336

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Military

*	MIL-A-2550	Ammunition and Special Weapons, General Specification for
*	MIL-C-6021	Castings, Classification and Inspection of
	MIL-H-6875	Heat Treatment of Steels (Aircraft Practice) Process for
*	MIL-M-7866	Molybdenum Disulfide Technical, Lubrication Grade
	MIL-W-8611	Welding, Metal Arc and Gas, Steels and Corrosion and Heat Resisting Alloys; Process for
*	MIL-R-11468	Radiographic Inspection; Soundness Requirements for Arc and Gas Welds in Steel
*	MIL-M-11473	Magnetic Particle Inspection: Soundness Requirements for Weldments
*	MIL-R-45226	Radiographic Inspection of Graphite

STANDARDS

Federal

	FED-STD-151	Metals, Test Method
*	FED-STD-595	Colors

Military

	MIL-STD-105	Sampling Procedures for Tables for Inspection by Attributes
*	MIL-STD-130	Identification Marking for U. S. Military Property

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*	MIL-STD-171	Finishing for Metal and Wood Surfaces
*	MIL-STD-453	Inspection, Radiographic

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

ARMY

MISSILE PURCHASE DESCRIPTIONS

MPD-561	Liner Compounds, Polysulfide, Preparation of
MPD-5010	Sealing Compound, Cold-Curing, Preparation of

DRAWINGS

Army Missile Command

8030871	Container, Shipping, Blast Tube and Nozzle Assembly (see 6.2)
8030880	Gas Generator Assembly, Metal Parts
8030881	Case and Adapter Assembly, Metal Parts
8034279	Container Assembly, Blast Tube and Nozzle, Rocket Motor, M30 (see 6.2)
8034291	Blast Tube and Nozzle Assembly
8034427	Container Assembly, Case and Adapter, Rocket Motor, M30

* (Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)
*

* 2.2 Other publications. The following document(s) forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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CODE OF FEDERAL REGULATIONS

*	49 CFR 100-199	Interstate Commerce Commissions, Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

* (Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC, 20402.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B311-58	Density of Cemented Carbides

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA, 19103.

* (Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Preproduction samples. Unless otherwise specified (see 6.2), the contractor shall furnish preproduction samples for preproduction inspection in accordance with 4.3. The preproduction samples shall be produced by the same methods, processes and techniques to be employed in subsequent production of the contract quantity.

3.2 Materials. Materials shall be in accordance with applicable drawings, specifications and drawings referenced thereon and as specified herein.

3.2. Carbon. The density of the carbon used for the nozzle insert shall be within the range of 2.18 to 2.27 grams per cubic centimeter.

3.2.2 Liner compound. Liner compound used for sealing the insulation to the adapter and blast tube shall conform to MPD-561, type II, class 7.

3.2.3 Sealing compound. Sealing compound used to secure the nozzle closure to the nozzle shall conform to MPD-5010.

3.3 Design. Parts and assemblies shall conform to dimensions, tolerance limits, physical properties and degree of surface roughness specified by applicable drawings and specifications referenced herein.

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3.3 Design. Parts and assemblies shall conform to dimensions, tolerance limits, physical properties and degree of surface roughness specified by applicable drawings and specifications referenced herein.

3.4 Construction. Parts and assemblies shall be constructed in accordance with Drawings 8030880, 8030881, 8034291, and applicable drawings and specifications listed thereon and as specified herein.

3.4.1 Alternate methods. The nozzle body (Drawing 8034198, covering a detail of Drawing 8034291) may be case or welded construction. The gas generator head (Drawing 8034202, covering a detail of Drawing 8030880) may be cast or machined from solid stock.

3.4.2 Welding. Welding shall be arc process in accordance with MIL-W-8611. External metal buildup shall be no greater than 0.075 inches. Internal metal buildup shall be no greater than 0.062 inches.

3.4.2.1 Pressure vessel. Soundness of welds in the pressure vessel shall be in accordance with MIL-R-11468, Standard I, except that voids up to 1/32 inch in diameter are allowable. Any 2 such voids less than one diameter apart shall be considered as one void. Any 2 voids within one inch of each other within the weld shall be cause for rejection.

3.4.2.2 Nonpressure vessel area. Soundness of welds in the nonpressure vessel area shall be in accordance with MIL-R-11468, Standard II.

3.4.3 Heat treatment. Parts requiring heat treatment shall be heat treated in accordance with MIL-H-6875 to the hardness ratings specified on applicable drawings and as specified in table I.

Table I - Yield Strengths

Components	Minimum psi	Maximum psi
Blast tube	60,000	140,000
Case Assembly	145,000	170,000
Adapter	110,000	

3.4.4 Adapter assembly. The mating surfaces of the adapter and adapter insulation (Drawing 8034187, covering a detail of Drawing 8030881) shall be sealed together with liner compound conforming to 3.2.2. After assembly, the compound shall be cured under an internal pressure of approximately 50 pounds per square inch (psi) in accordance with the specification covering the compound.

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3.4.5 Blast tube and nozzle assembly.

3.4.5.1 Blast tube assembly. The mating surfaces of the blast tube and blast tube insulaton shall be sealed together with liner compound conforming to 3.2.2. After assembly, the compound shall be cured in accordance with the specification covering the compound. Approximately 100 psi pressure shall be applied at locations specified on Drawing 8034192 (covering a detail of Drawing 8034291) during cure of liner compound.

3.4.5.2 Blast tube and nozzle coupling. The blast tube and nozzle shall be securely coupled using coupling (Drawing 8034196, covering a detail of Drawing 8034291). Molybdenum disulfide powder lubricant conforming to MIL-M-7866 shall be applied to 15 degree angle surface prior to installation of the coupling. The coupling shall be torqued to 35 foot pounds.

3.4.5.2.1 Nozzle closure. The nozzle closure (Drawing 8034200, covering a detail of Drawing 8034291) shall be sealed securely to the nozzle using sealing compound conforming to 3.2.3. The compound shall be applied uniformly to the entire periphery of the nozzle closure.

3.4.5.2.2 Nozzle blanket insulation. The nozzle blanket insulation (Drawing 8034329, covering a detail of Drawing 8034291) shall be mounted in such a position that the insulation interfaces fit snugly together. The insulation shall be secured in accordance with Drawing 8034291 using wire .032 inch in diameter conforming to QQ-W-423, Form 1, Composition FS 302 or 304, Condition A.

3.5 Performance and product characteristics.

3.5.1 Cast nozzle body. After machining, the cast nozzle interior surface adjacent to the carbon insert for a distance of .750 inch aft shall not contain more than five (5) surface craters or pits greater than .125 inch diameter by .050 inch deep with a minimum separation of .750 inch. From a girth line two (2) inches aft of the carbon insert to aft end of nozzle, the interior surface shall not contain more than thirty (30) surface craters or pits greater than .375 inch in length by .125 inch in width by .030 inch in depth with a minimum separation of .750 inch. There shall be no more than nine (9) defects within nine (9) square inches. The total allowable defects on the exterior surface of the nozzle shall be no more than thirty-two (32), each shall be no greater than .375 inch in length by .0625 inch in width by .035 inch in depth, and there shall be not more than eleven (11) in any area of twelve (12) square inches. Craters or pits are not permitted on gasket, sealing of flange mating surfaces.

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3.5.2 Nozzle carbon insert. After final machining of the carbon insert, the surface in the throat, and 0.50 inch on either side, shall have no chips, flakes or flaws exceeding 0.015 inch in depth or .625 inch in length. The remaining interior surface shall have no chips, flakes or flaws greater than 0.030 inch in depth or 0.1875 inch in length. The exterior surface of the insert shall have no defects greater than .09375 inch diameter and .030 in depth and no defects are permitted within 0.50 inch of either end. The insert shall have no internal cracks. There shall be no voids over .09375 inch in diameter or voids totaling a volume equivalent to a .21875 inch diameter sphere in a one inch cube.

3.5.3 Hydrostatic pressure.

* 3.5.3.1 Blast tube, case assembly, adapter. The blast tube (Drawing 8034193 covering a detail of Drawing 8034291), case assembly (Drawing 8034172, covering a detail of Drawing 8030881), adapter (Drawing 8034187, covering a detail of Drawing 8030881), shall withstand an internal hydrostatic pressure of 900 psi for a period of 5 minutes without leakage or permanent distortion in excess of drawing dimensions.

3.5.3.2 Nozzle body. After machining, the nozzle body (Drawing 8034198, covering a detail of Drawing 8034291) shall withstand an internal hydrostatic pressure of 600 psi for a period of 3 minutes without rupture or permanent distortion in excess of drawing dimensions.

3.5.4 Weight and balance. The weight of the completed blast tube and nozzle assembly shall be 184.0, plus or minus 5.4, pounds. The CG shall be 49.37, plus or minus 0.72, inches aft of the forward face of the blast tube.

3.5.5 Blast tube insulation. There shall be no visible cavities exceeding 0.062 inches in any dimension in the blast tube insulation.

3.6 Pretreatment and protective finish. Unless otherwise specified on detail drawings, preparation, pretreatment and finish of metal surfaces shall be as follows:

3.6.1 Pretreatment. All metal surfaces shall be cleaned and treated prior to finishing in accordance with MIL-STD-171, Finish No. 4.2 and 5.1. Stock removed from all surfaces during this operation shall be not greater than .001 inch.

3.6.2 Protective finish. A protective coating of primer, conforming to Desoto Chemical, Inc., Koropon No. 5210 CA, Redoxide Primer, with No. 9350 Koropon CA Activator, Thinner No. S-4800 Koropon Solvent, or equal, shall be applied to all exterior metal surfaces and the interior surface of the nozzle assembly (Drawing 8034197, covering a detail of Drawing 8034291) from the aft end of the carbon insert to aft end of nozzle, and the open end of the gas generator nozzle (Drawing 8034213, covering a detail of Drawing 8030880) except as follows:

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3.6.2.1 Threads and mating surfaces. Threads and mating surfaces shall be kept free of finishing materials.

3.6.2.2 Ring, pins and plugs. All surfaces of the ring (Drawing 8034218, covering a detail of Drawing 8030881), pins (Drawing 8034185, covering a detail of Drawing 8030881) and gas generator plugs (Drawing 8034214, covering a detail of Drawing 8030880) shall be kept free of finish materials.

3.6.3 Final finish. Two coats of enamel conforming to TT-E-516. Color X34087 specified in FED-STD-595 shall be applied on the exterior surface of the blast tube assembly (Drawing 8034192, covering a detail of Drawing 8034291), the exterior surface of the nozzle assembly (Drawing 8034197, covering a detail of Drawing 8034291) and on the interior surface from the aft face of the installed nozzle closure to the aft end of the nozzle.

3.7 Marking.

3.7.1 Production marking. Parts and assemblies shall be marked for identification in accordance with MIL-STD-130 and shall be located as specified on applicable drawings.

3.7.2 Special marking. Upon satisfactory completion of hydrostatic tests, parts listed in 3.5.3 shall be marked with the letter "H" and shall be impressed in the location specified on respective drawings. The CG of each blast tube and nozzle assembly, Drawing 8034291, shall be marked by a circumferential band 1 2, plus or minus 0.03, inches wide centered on the actual longitudinal CG. The band may be painted using Color No. 31136 conforming to FED-STD-595 or applied by decalcomania transfer. The weight, CG and the note "PLACE HANDLING STRAPS HERE" shall be printed or stencilled on the band.

3.8 Workmanship. Parts and assemblies shall be free of dirt, grease, chips, rust or other foreign matter. There shall be no cracks, bends, breaks, dents, grooves or other defects which would make any part or assembly unsuitable for the intended purpose. The workmanship shall be of such quality that will insure uniformity of performance and interchangeability of parts and assemblies having the same part numbers.

4. QUALITY ASSURANCE PROVISIONS

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* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Lot Size. A lot of metal parts for M30 Rocket Motors shall consist of that quantity produced in not more than 24 hours of continuous production with no change in facilities, process or specifications which would change the chemical and physical properties or interchangeability of the parts.

4.3 Preproduction provisions.

4.3.1 Sample. Unless otherwise specified, (see 6.2) the sample shall consist of one complete assembly and one complete set of metal parts.

4.3.2 Inspection location. Preproduction inspection shall be performed by the Government at a place specified by the procuring activity (see 6.2).

4.3.3 Inspection. The sample specified in 4.3.1 shall be subjected to the examination and tests specified in 4.4.2.

4.4 Production provisions.

4.4.1 Sampling.

4.4.1.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105 at inspection level II.

4.4.1.2 Test samples. Test samples shall be comprised for all welded parts and assemblies produced and all material used for the carbon inserts.

4.4.2 Inspections.

4.4.2.1 Examination. Samples selected in 4.4.1.1 shall be examined for conformance to requirements specified herein.

4.4.2.2 Acceptance examination. Preservation, packing and marking shall be examined to determine conformance to requirements in section 5.

4.4.2.3 Testing. Samples selected in 4.4.1.2 shall be subjected to the tests as specified in 4.4.3.

4.4.3 Tests.

4.4.3.1 Carbon. The density of the material (see 3.2.1) shall be determined in accordance with ASTM B311-58.

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4.4.3.2 Radiographic inspection.

* 4.4.3.2.1 Weldments. Radiographic inspection of the weldments shall be conducted in accordance with MIL-STD-453.

4.4.3.2.1.1 Nonpressure vessel area. Soundness of welds in the nonpressure vessel area shall be determined in accordance with MIL-R-11468.

4.4.3.2.2 Castings. The cast nozzle (Drawing 8034198, covering a detail of Drawing 8034291) and the gas generator cast head (Drawing 8034202, covering a detail of Drawing 8030880) shall be inspected in accordance with MIL-C-6021 after rough machining.

* 4.4.3.2.3 Nozzel carbon insert. Each carbon insert produced shall be subjected to radiographic inspection in accordance with MIL-R-45226 to determine conformance to 3.5.2.

4.4.3.3 Magnetic particle. All weldments of each rocket motor assembly and each cast nozzle produced shall be magnetic particle inspected in accordance with MIL-M-11473 after hydrostatic test.

4.4.3.4 Rockwell hardness. Rockwell hardness of heat treated parts and assemblies shall be determined in accordance with FED-STD-151 to determine conformance to table I.

4.4.3.5 Hydrostatic test.

4.4.3.5.1 Blast tube, case assembly, adapter. After completion of fabrication, including heat treatment and final machining, the blast tube, case assembly and adapter shall be tested to determine compliance with 3.5.3.1. The tests shall be such that the components are free to expand in any direction during pressurization. At least 48 hours shall elapse between heat treatment and hydrostatic test. The test may, at the option of the contractor, be performed with the components assembled as one unit or with the blast tube separately.

4.4.3.5.2 Nozzle. Each blast tube nozzle, cast or welded construction, shall be tested to determine compliance with 3.5.3.2. The tests shall be such that the nozzle is free to expand in any direction during pressurization. The cast nozzle shall be normalized prior to testing.

* 5. PACKAGING

* 5.1 Preservation, packing, unitization, and marking. Level A, B or Commercial Packaging (see 6.2)

* 5.1.1 Level A or B. Preservation, packing, unitization and marking shall be in

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accordance with packaging data sheets APN 8034427, 8034279 or 8030871 and specification referenced therein and Interstate Commerce Commission requirements specified in Code of Federal Regulations 49 CFR 100-199.

* 5.1.2 Commercial. Not applicable.

* 5.2 Data cards. Data cards shall be provided in accordance with MIL-A-2550.

6. NOTES

6.1 Intended use. These metal parts are intended for use in Rocket Motor M30, used as the sustainer motor in Guided Missile, Surface-to-Air, Nike-Hercules.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Whether a preproduction sample is required (see 3.1 and 4.3.1).

(c) Inspection location (see 4.3.2)

* (d) Type shipping container to be used for blast tube and nozzle assembly (see 2.1 and 5.1.1)

* (e) Packaging requirements if other than specified (see 5.1)

* (f) Data cards in accordance with MIL-A-2550.

* 6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* CUSTODIAN:
Army-MI

Preparing activity:
Army-MI

Project No. 1336-A328

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DOCUMENT IDENTIFIER AND TITLE

MIL-R-46509A(MI) ROCKET MOTOR, M30, METAL PARTS FOR

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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